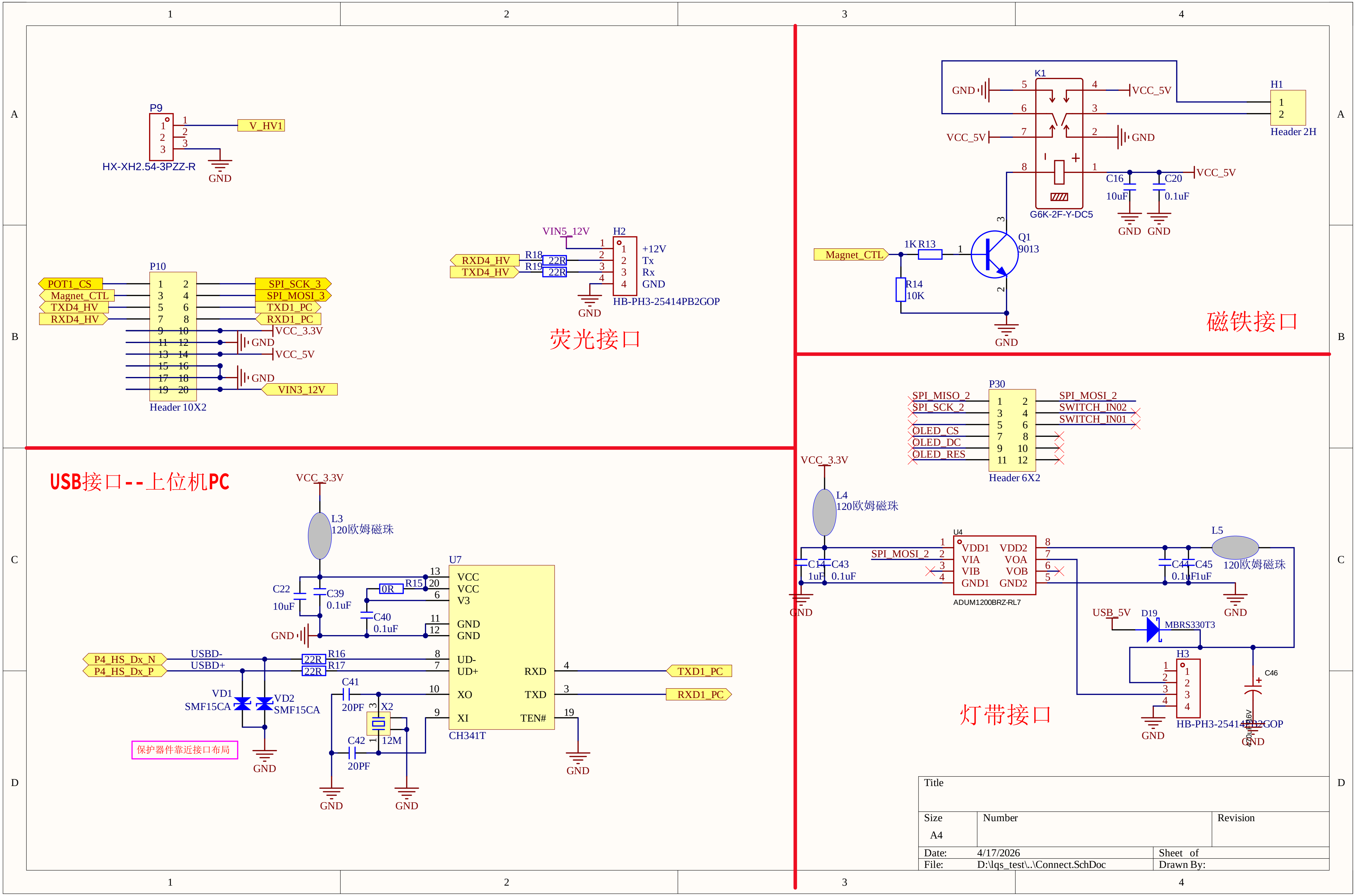

Title

Size
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Number

Revision

Date: 4/17/2026

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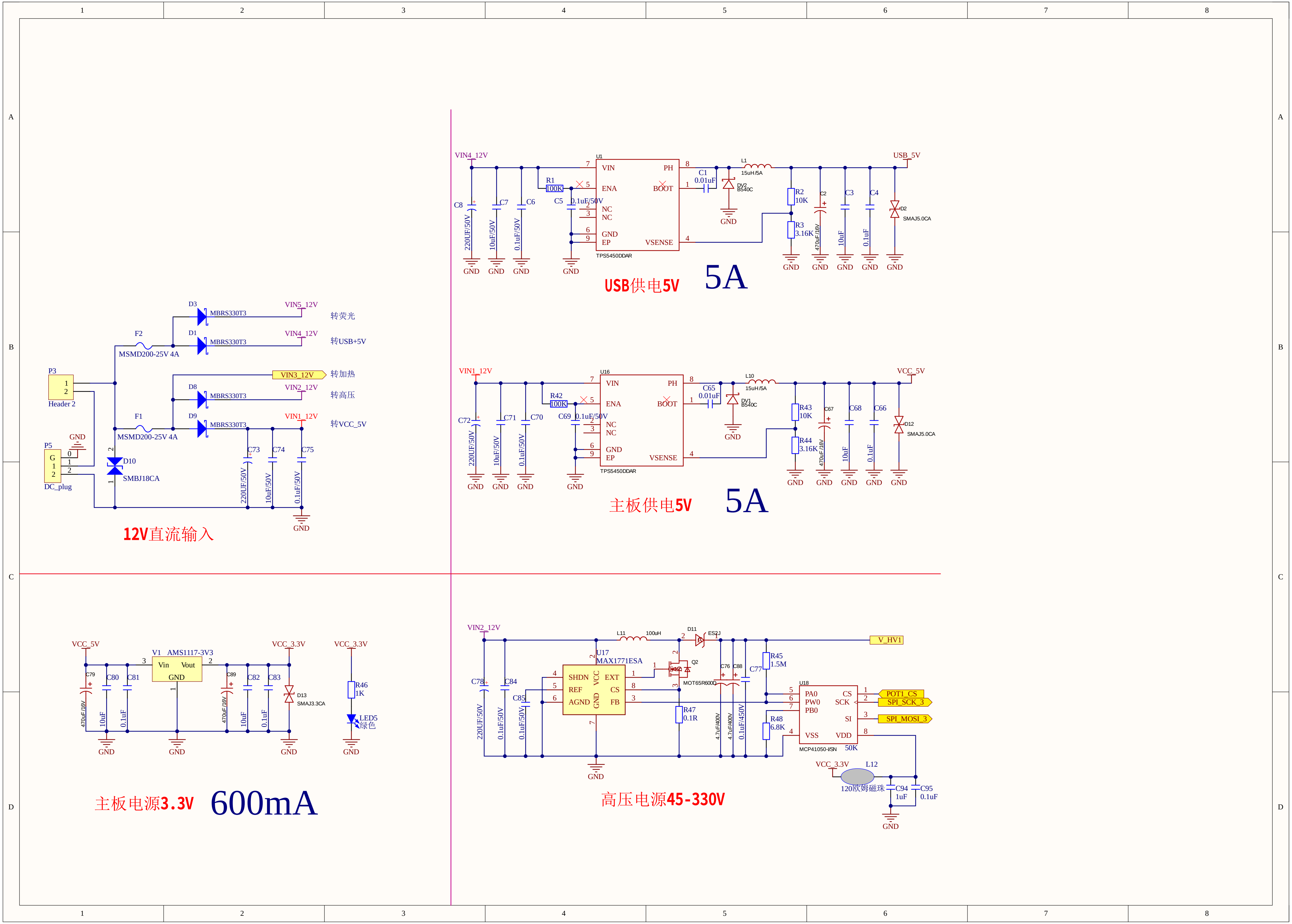
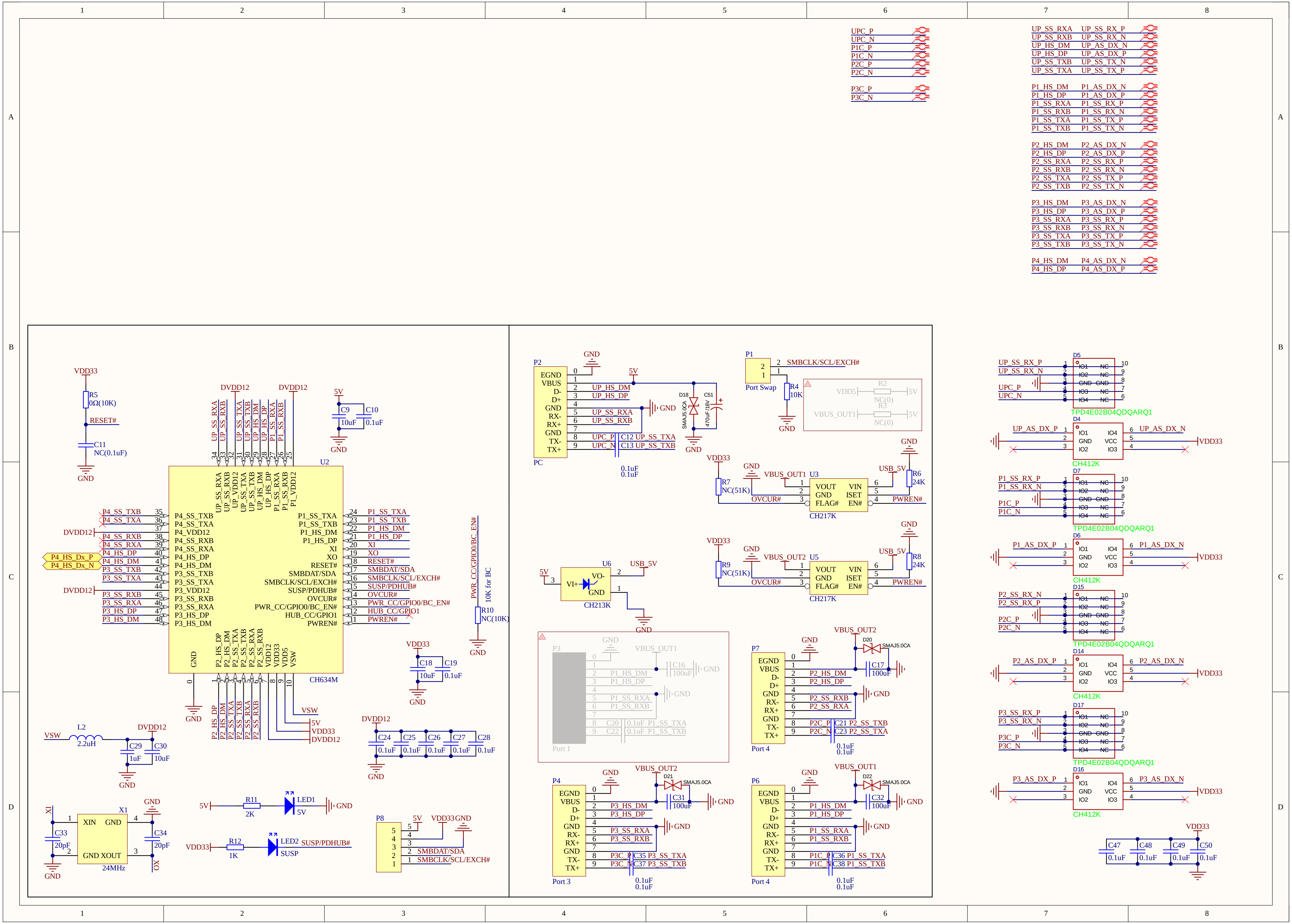
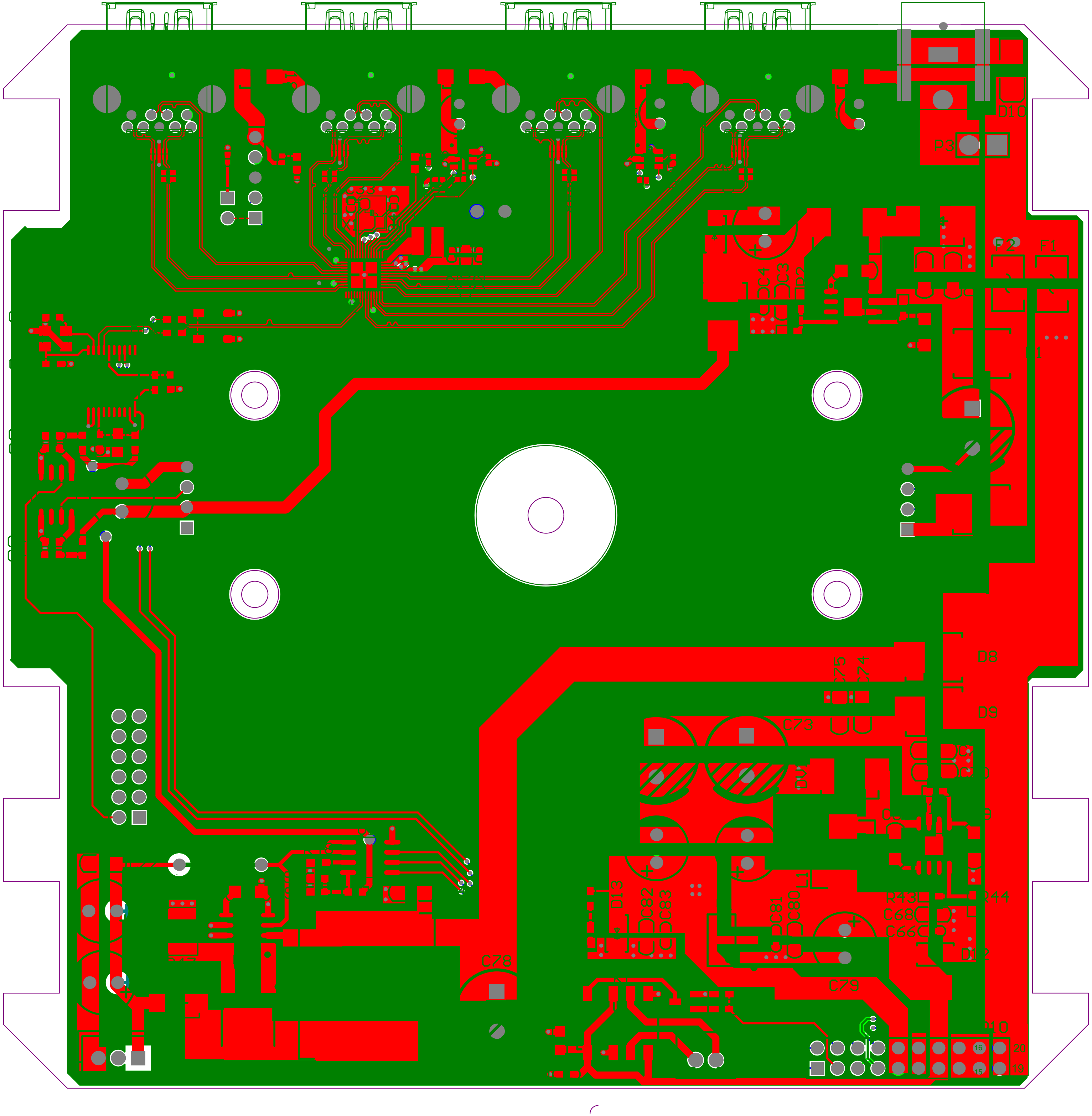

USB供电5V
5A
VIN4_12V
U1
TPS5450DDAR
VIN
ENA
NC
NC
GND
EP
PH
BOOT
VSENSE
R1 100K
C5 0.1uF/50V
C8 220UF/50V
C7 10uF/50V
C6 0.1uF/50V
C1 0.01uF
DV2 B540C
L1 15uH/5A
R2 10K
R3 3.16K
C2 470uF/16V
C3 10uF
C4 0.1uF
D2 SMAJ5.0CA
USB_5V
GND
主板供电5V
5A
VIN1_12V
U16
TPS5450DDAR
R42 100K
C69 0.1uF/50V
C72 220UF/50V
C71 10uF/50V
C70 0.1uF/50V
C65 0.01uF
DV1 B540C
L10 15uH/5A
R43 10K
R44 3.16K
C67 470uF/16V
C68 10uF
C66 0.1uF
D12 SMAJ5.0CA
VCC_5V
12V直流输入
P3
Header 2
P5
DC_plug
G
F2 MSMD200-25V 4A
F1 MSMD200-25V 4A
D10 SMBJ18CA
D3 MBRS330T3
D1 MBRS330T3
D8 MBRS330T3
D9 MBRS330T3
VIN5_12V 转荧光
VIN4_12V 转USB+5V
VIN3_12V 转加热
VIN2_12V 转高压
VIN1_12V 转VCC_5V
C73 220UF/50V
C74 10uF/50V
C75 0.1uF/50V
主板电源3.3V
600mA
VCC_5V
V1 AMS1117-3V3
Vin
Vout
GND
C79 470uF/16V
C80 10uF
C81 0.1uF
C89 470uF/16V
C82 10uF
C83 0.1uF
D13 SMAJ3.3CA
VCC_3.3V
R46 1K
LED5 绿色
高压电源45-330V
VIN2_12V
L11 100uH
D11 ES2J
V_HV1
U17 MAX1771ESA
SHDN
REF
AGND
VCC
GND
EXT
CS
FB
Q2 MOT65R600Q
R47 0.1R
C78 220UF/50V
C84 0.1uF/50V
C85 0.1uF/50V
C76 4.7uF/400V
C88 4.7uF/400V
C77 0.1uF/450V
R45 1.5M
R48 6.8K
U18 MCP41050-I/SN
PA0
PW0
PB0
VSS
CS
SCK
SI
VDD
50K
POT1_CS
SPI_SCK_3
SPI_MOSI_3
VCC_3.3V
L12
120欧姆磁珠
C94 1uF
C95 0.1uF
GND


UPC_P
UPC_N
P1C_P
P1C_N
P2C_P
P2C_N
P3C_P
P3C_N
UP_SS_RXA UP_SS_RX_P
UP_SS_RXB UP_SS_RX_N
UP_HS_DM UP_AS_DX_N
UP_HS_DP UP_AS_DX_P
UP_SS_TXB UP_SS_TX_N
UP_SS_TXA UP_SS_TX_P
P1_HS_DM P1_AS_DX_N
P1_HS_DP P1_AS_DX_P
P1_SS_RXA P1_SS_RX_P
P1_SS_RXB P1_SS_RX_N
P1_SS_TXA P1_SS_TX_P
P1_SS_TXB P1_SS_TX_N
P2_HS_DM P2_AS_DX_N
P2_HS_DP P2_AS_DX_P
P2_SS_RXA P2_SS_RX_P
P2_SS_RXB P2_SS_RX_N
P2_SS_TXA P2_SS_TX_P
P2_SS_TXB P2_SS_TX_N
P3_HS_DM P3_AS_DX_N
P3_HS_DP P3_AS_DX_P
P3_SS_RXA P3_SS_RX_P
P3_SS_RXB P3_SS_RX_N
P3_SS_TXA P3_SS_TX_P
P3_SS_TXB P3_SS_TX_N
P4_HS_DM P4_AS_DX_N
P4_HS_DP P4_AS_DX_P
VDD33
R5
0Ω(10K)
RESET#
C11
NC(0.1uF)
GND
DVDD12
5V
C9 10uF
C10 0.1uF
U2
CH634M
P4_SS_TXB
P4_SS_TXA
P4_VDD12
P4_SS_RXB
P4_SS_RXA
P4_HS_DP
P4_HS_DM
P3_SS_TXB
P3_SS_TXA
P3_VDD12
P3_SS_RXB
P3_SS_RXA
P3_HS_DP
P3_HS_DM
P1_SS_TXA
P1_SS_TXB
P1_HS_DM
P1_HS_DP
XI
XO
RESET#
SMBDAT/SDA
SMBCLK/SCL/EXCH#
SUSP/PDHUB#
OVCUR#
PWR_CC/GPIO0/BC_EN#
HUB_CC/GPIO1
PWREN#
P4_HS_Dx_P
P4_HS_Dx_N
P2_HS_DP
P2_HS_DM
P2_SS_TXA
P2_SS_TXB
P2_SS_RXA
P2_SS_RXB
DVDD12
VDD33
VDD5
VSW
PWR_CC/GPIO0/BC_EN#
10K for BC
R10
NC(10K)
C18 10uF
C19 0.1uF
L2 2.2uH
C29 1uF
C30 10uF
C24 C25 C26 C27 C28 0.1uF
X1 24MHz
XIN GND XOUT
C33 20pF
C34 20pF
R11 2K
LED1 5V
R12 1K
LED2 SUSP
SUSP/PDHUB#
P8
SMBDAT/SDA
SMBCLK/SCL/EXCH#
P2
PC
EGND VBUS D- D+ GND RX- RX+ GND TX- TX+
UP_HS_DM
UP_HS_DP
UP_SS_RXA
UP_SS_RXB
C12 UP_SS_TXA
C13 UP_SS_TXB
0.1uF
D18 SMAJ5.0CA
C51 470uF/16V
P1
Port Swap
R4 10K
R2 VDD5 NC(0) 5V
R3 VBUS_OUT1 NC(0) 5V
U3 CH217K
U5 CH217K
VOUT VIN GND ISET FLAG# EN#
R7 NC(51K)
R9 NC(51K)
R6 24K
R8 24K
USB_5V
VBUS_OUT1
VBUS_OUT2
U6 CH213K
VI+ VO- GND
P3 Port 1
C16 100uF
C20 C22 0.1uF P1_SS_TXA P1_SS_TXB
P7 Port 4
D20 SMAJ5.0CA
C17 100uF
C21 P2_SS_TXB
C23 P2_SS_TXA
P4 Port 3
D21 SMAJ5.0CA
C31 100uF
C35 P3_SS_TXA
C37 P3_SS_TXB
P6 Port 4
D22 SMAJ5.0CA
C32 100uF
C36 P1_SS_TXA
C38 P1_SS_TXB
D5 D7 D15 D17 TPD4E02B04QDQARQ1
IO1 IO2 GND IO3 IO4 NC
D4 D6 D14 D16 CH412K
IO1 GND IO2 IO4 VCC IO3
C47 C48 C49 C50 0.1uF


D10
P3
F2
F1
C4
C3
D8
D9
C73
C75
C74
C78
C79
D13
C82
C83
C81
C80
L1
R43
R44
C68
C66
D12
P10
16
15
20
19